

SOME SCHEMES OF REALIZATION DIGITAL FIR FILTERS ON FPGA FOR COMMUNICATION APPLICATIONS

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Abstract — Realization of digital Finite-Impulse-Response (FIR) filters are so important issues for digital communications and signal processing applications which is the main contribution of this paper. The direct and transposed architectures are employed to implement FIR filters on field-programmable-gate-array (FPGA) using a single SPARTAN 2 chip, XC2S50-5I-tq144, from Xilinx, Inc. The Verilog hardware description language is used for designing different schemes of FIR filter. The effects of coefficient symmetry and implementing multiplier blocks by shift and add operators on frequency and FPGA chip area occupation metrics are studied. Simulation results demonstrate improving FPGA-based solutions for digital FIR filter design.

I. Introduction

Nowadays, the field programmable gate array (FPGA) technology is quit mature for digital signal processing (DSP) applications [1] due to fast progress in very large scale integration (VLSI) technology. The FPGA devices [2] provide fully programmable system-on-chip environments by incorporating the programmability of programmable logic devices and the architecture of gate arrays. They consist of thousands of logic gates and some configurable logic blocks which make them an appropriate solution for prototyping the application specific integrated circuits with dedicated architectures for specified DSP applications.

The digital Finite-Impulse-Response (FIR) filters are mainly employed in digital signal processing applications. Design and implementation of digital FIR filters have received much attention by researchers in literature. There are two mainly designing environments for implementing digital FIR filters; digital signal processor (DSP)-based and FPGA-based solutions. Unlike traditional solutions based on microprocessors, microcontrollers and DSP devices, the FPGAs provide design flexibility and hardware parallelism by breaking the paradigm of sequential execution for high speed applications. Daitx et al. [3] proposed a VHDL approach for designing optimized FIR filters. They used general coefficient representation for the optimal sharing of partial products in multiple constants multiplications. A.A.H. Ab-Rahman et al. [4] presented ASIC design flow for the implementation of adaptive FIR filter using MATLAB and Mentor Graphics IC Design tools. Razak et al. [5] described the design of transposed form FIR filter on FPGAs using VHDL. In [6] the design optimization of one and two dimensional fully pipelined computing structures for area-delay-power-efficient implementation of FIR filter by systolic decomposition of distributed arithmetic based inner product computation was proposed.

The main components of digital FIR filters design on FPGAs are the register bank to save the samples of signals, adder to implement sum operations and multiplier for multiplication the filter coefficients to signal samples. Although, design and implementation of digital FIR filters seem simple but the design bottleneck is multiplier block for speed, power consumption and FPGA chip area occupation. A common interesting idea in digital circuits design is replacing multiplier block by some shift and add operations and sharing common terms.

This will significantly reduces the complexity of implementing multiplier block and consequently digital filter design. In this paper design and implementing of digital FIR filters on FPGA is considered. The common methods called direct and transposed architectures are employed for designing FIR filters. The effects of coefficient symmetry and replacing multiplier blocks by shift and add operators on frequency and chip area occupation are evaluated. This is important to choose the proper designing approach for specific applications.

The rest of the paper is organized as follows. Section II presents the FIR filter design. The different implementing architectures of FIR filter are described in section III. The simulation results achieved by ISE for each module are presented in section IV. Finally the paper is concluded in section V.

II. Digital FIR Filters

The digital FIR filters with constant coefficients are linear time invariant. The response of an N-order digital FIR filter to input $X[n]$ is a convolution sum that could be written as follows.

$$Y[n] = X[n] * H[n] = \sum_{k=0}^{N-1} H[k] X[n-k] \quad (1)$$

where $H[k]$ is the filter coefficients.

Design of FIR filter for a specific application includes calculating the coefficients according to different criteria like filter order, sampling frequency, pass-band and stop-band frequencies, etc. The coefficient calculating could be done by different software which makes it as simple task. The implementing of FIR filters is a challenges problem due to constraints on power, delay and area in digital circuits design. Therefore, different approaches were introduced in literature which are mainly based on direct and transposed architectures.

The direct form of an LTI FIR filter is shown in figure 1 which mainly consists of shift registers, adders and multipliers. The signal samples are multiplied by filter coefficients and are gathered together in adder block. The transposed structure, a modification of direct form, of FIR filter is shown in figure 2. As shown, in transposed form the shift register is not used which makes it simpler for implementation than direct form. The coefficients of a typical low-pass FIR filter were computed using Matlab FDA toolbox [7]. This filter is used as a test bench for evaluating algorithms performances for different applications. The parameters of the proposed low-pass FIR filter are summarized in table 1. The calculated coefficients of the proposed filter are $H = [-7, 37, 63, -7, 37]$. The coefficients are symmetric around the center, 63, which is used for reducing design complexity.

III. Implementation Architectures

Design This section describes different architectures for implementing digital FIR filters on FPGA. The first structure, shown in figure 3, is based on direct architecture, figure 1, which includes some multiplier blocks. In the proposed 5-length FIR filter the coefficients are

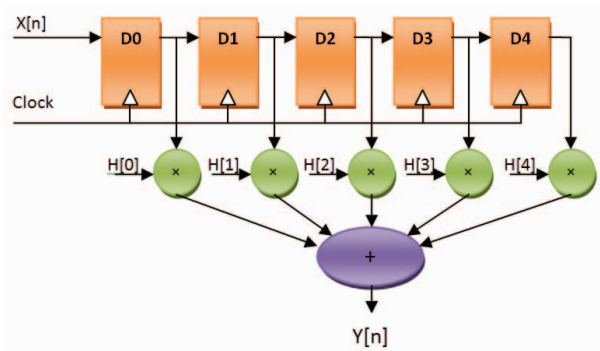


Figure 1. The direct architecture of a typical FIR filter

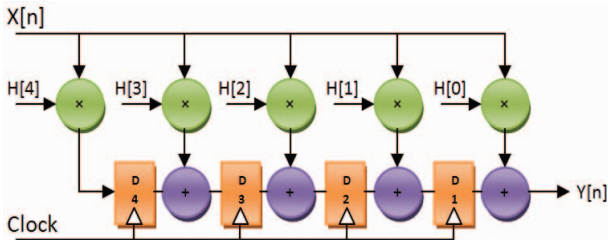


Figure 2. The transposed FIR filter architecture

Table 1. The parameter of low pass FIR filter

Filter Parameter	Value
Design Method	Equiripple
Order	4
Density Factor	20
Sampling Frequency	$F_s = 2\text{MHz}$
Passband Frequency	$F_{\text{pass}} = 500\text{KHz}$
Stopband Frequency	$F_{\text{stop}} = 700\text{KHz}$
Passband Weight	$W_{\text{pass}} = 0.1$
Stopband Weight	$W_{\text{stop}} = 0.1$

symmetric around the center, so the number of multipliers could be reduced to 3 multipliers instead of 5 multipliers and the number of adders is increased to 3 adders. Therefore, the FPGA chip area occupation is significantly reduced. The aim of second algorithm is to implement the multiplier blocks by shift and add operators. The coefficients are encoded into canonical signed digit as follows.

$$-7 = -(111) = -(100\bar{1}) \quad (2a)$$

$$37 = 100101 \quad (2b)$$

$$63 = 111111 = (100000\bar{1}) \quad (2c)$$

As the input of all multipliers is the same, the higher order shifts could be implemented using the results of lower order shifts. For example, the $X[n] \ll 3$ sub-operation (3-order left shift of $X[n]$) could be done by one step left shifting of $X[n] \ll 2$. The implementing of multiplier blocks are described as follows using the canonical signed digit form of coefficients.

$$X[n] * -7 = -X[n] \ll 3 + X[n]$$

$$X[n] * 37 = X[n] \ll 5 + X[n] \ll 2 + X[n]$$

$$X[n] * 63 = X[n] \ll 6 - X[n]$$

The filter structure is shown in figure 4. As shown, the structure is multiplier-less and consequently the filter design complexity would significantly decrease.

The third algorithm is based on transposed architecture where the number of multiplier blocks is reduced due to coefficient symmetry, shown in figure 5. The fourth algorithm is also used transposed form. It has no multiplier by employing canonical signed digit form of coefficients and implementing them by shift operators, figure 6.

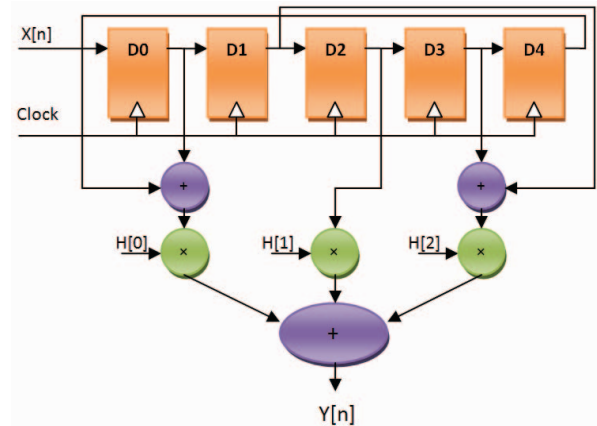


Figure 3. The symmetric direct form of FIR filter

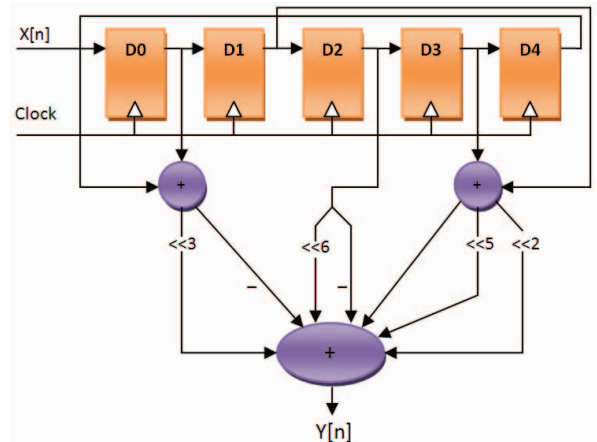


Figure 4. Multiplier-less direct form of FIR filter

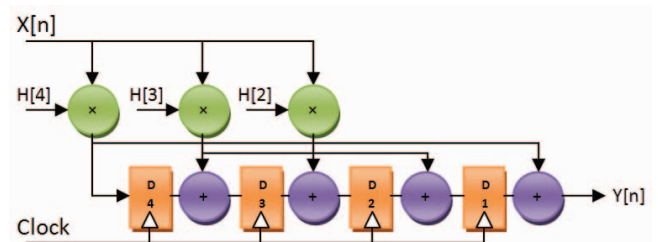


Figure 5. The symmetric transposed form of FIR filter

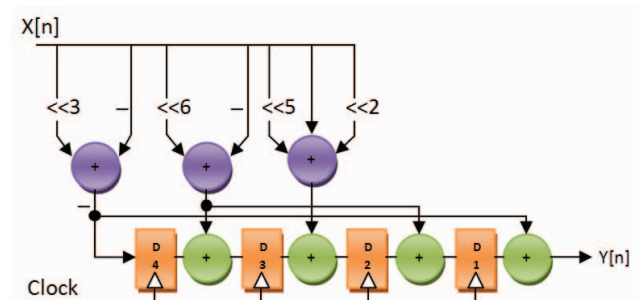


Figure 6. Multiplier-less transposed form of FIR filter

IV. Results

This section presents the simulation results of implementation the digital FIR filter on single FPGA chip using different architectures. The FIR filter is implemented on SPARTAN 2; chip number XC2S50-5I-tq144, from Xilinx Inc [8]. The characteristics of proposed FPGA chip are summarized in table 2. The different architectures of proposed FIR filter are implemented by Verilog language and the Xilinx ISE 10.1 Web-Pack edition is used for synthesize and simulation the Verilog codes. The frequency response and the impulse response of proposed filter are interpreted in figure 7 and figure 8, respectively.

Table 2. The characteristic of FPGA chip

System Gates(Logic and RAM)	50000
Slices	768
Flip Flops	1536
LUTs	1536
IOBs	92
GCLK	4

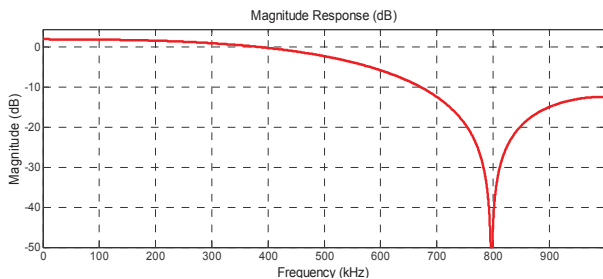


Figure 7. The frequency response of FIR filter

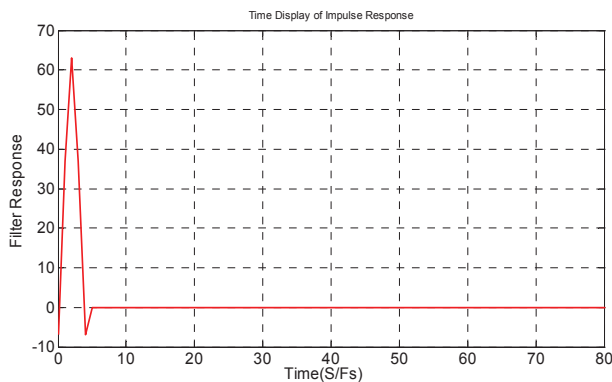


Figure 8. The impulse response of proposed FIR Filter

Figure 9 shows the simulation results of synthesized FIR filter on FPGA. As shown the response of synthesized filter on FPGA is the same as simulated filter by MATLAB simulation FDA toolbox in figure 8. The performance of four synthesized architectures on FPGA is compared together according to utilized FPGA slices, flip flops and LUTs. The frequency of architectures is also

studied for fair decision making. The results are summarized in table 3. As shown, the direct architecture, for both symmetric and multiplier-less cases, uses flip flop less than the transposed architecture. The symmetric and multiplier-less transposed forms uses 43.75% and 9.4% flip-flops more than the same direct schemes, respectively. The number of used slices by direct architecture is more than that for transposed architecture for both symmetric and multiplier-less cases. The symmetric and multiplier-less direct forms uses 27% and 95% slices more than symmetric and multiplier-less transposed forms. For LUTs utilization, the number of used LUTs by direct architecture is more than that for transposed architecture for both symmetric and multiplier-less cases. The symmetric and multiplier-less direct forms uses 34.8% and 97.1% LUTs more than symmetric and multiplier-less direct forms, respectively.

The maximum operating frequency of synthesized architectures is given in table 4. The behavior of direct architecture is better than transposed architecture for this metric. The multiplier-less direct form has the highest operating frequency, 250.313 MHz, and the symmetric transposed form has the lowest operating frequency, 160.462 MHz. Furthermore, it could be concluded from the results that the multiplier-less structure has better performance than symmetric structure in both architecture for delay sensitive applications.

Table 3. The used space of FPGA chip

Design Approach	Flip Flops	Slices	LUTs
Symmetric Direct	64	197	372
Multiplier-less Direct	64	162	272
Symmetric Transposed	92	155	276
Multiplier-less Transposed	70	83	138

Table 4. The operating frequency of synthesized architectures on FPGA

Design Approach	Frequency(MHZ)
Symmetric Direct	240.096
Multiplier-less Direct	250.313
Symmetric Transposed	160.462
Multiplier-less Transposed	168.577

V. Conclusions

The realization of digital FIR filters on field programmable gate array devices was proposed. Two common architectures called direct and transposed architectures were employed for implementing FIR filters on a Xilinx SPARTAN2- XC2S50-5I-tq144 FPGA using Verilog hardware description language codes. The effects of coefficient symmetry and implementing multiplier blocks by shift and add operators on key metrics including number of utilized flip-flops, slices, and LUTs and maximum operating frequency were studied. The results

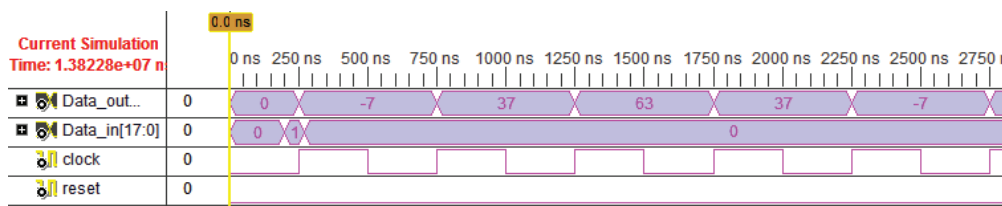


Figure 9. The impulse response of implemented FIR filter on FPGA

demonstrate that the transposed architecture uses FPGA area less than direct form. While the direct form has better performance from maximum usable frequencies view point. Furthermore, multiplier-less approaches improve FPGA utilization area metric more than coefficient symmetry. The later has better frequency performance.

VI. References

- [1] *Baese U. M.*, Springer, 2007.
- [2] *Wolf W.*, Englewood Cliffs, Prentice- Hall, 2004.
- [3] *Daitx F.F. et al.*, in Proceeding of International Conference on Signals, Circuits and Systems, pp. 1-5, 2008.
- [4] *Ab-Rahman A.A.H. et al.*, in Proceeding of International Conference on Electronic Design, Penang, Malaysia, pp. 1-4, Dec. 2008.
- [5] *Razak A.H.A. et al.*, in Proceeding of Asia-Pacific Conference on Applied Electromagnetic, Melaka, Malaysia, pp. 1-5, Dec. 2007.
- [6] *Meher P. K. et al.*, IEEE Tran. on Sig. Pro., vol. 56, no. 7, pp. 3009-3017, 2008.
- [7] *www.mathworks.com*
- [8] *www.xilinx.com*